

MLR

October 31, 2025

```
[1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import KFold, cross_val_score
from sklearn.metrics import mean_squared_error, mean_absolute_error, r2_score

[5]: X_train = pd.read_csv("X_train.csv")
X_test = pd.read_csv("X_test.csv")
y_train = pd.read_csv("y_train.csv").squeeze() # squeeze to convert to Series
y_test = pd.read_csv("y_test.csv").squeeze()

[6]: mlr_model = LinearRegression()
mlr_model.fit(X_train, y_train)

# =====
# K-Fold Cross Validation (k=10)
# =====
kf = KFold(n_splits=10, shuffle=True, random_state=42)

cv_r2 = cross_val_score(mlr_model, X_train, y_train, cv=kf, scoring='r2')
cv_rmse = np.sqrt(-cross_val_score(mlr_model, X_train, y_train, cv=kf,
    ↪scoring='neg_mean_squared_error'))
cv_mae = -cross_val_score(mlr_model, X_train, y_train, cv=kf,
    ↪scoring='neg_mean_absolute_error')

print("\n Cross-validation results (10-fold):")
print(f"R2 → mean: {cv_r2.mean():.3f}, std: {cv_r2.std():.3f}")
print(f"RMSE → mean: {cv_rmse.mean():.3f}, std: {cv_rmse.std():.3f}")
print(f"MAE → mean: {cv_mae.mean():.3f}, std: {cv_mae.std():.3f}")

# =====
# Evaluate on Test Set
# =====
y_pred = mlr_model.predict(X_test)
```

```

r2 = r2_score(y_test, y_pred)
rmse = np.sqrt(mean_squared_error(y_test, y_pred))
mae = mean_absolute_error(y_test, y_pred)

print("\n Test Set Performance:")
print(f"R2    = {r2:.3f}")
print(f"RMSE = {rmse:.3f}")
print(f"MAE  = {mae:.3f}")

```

Cross-validation results (10-fold):

R² → mean: 0.476, std: 0.081

RMSE → mean: 1.055, std: 0.074

MAE → mean: 0.842, std: 0.045

Test Set Performance:

R² = 0.501

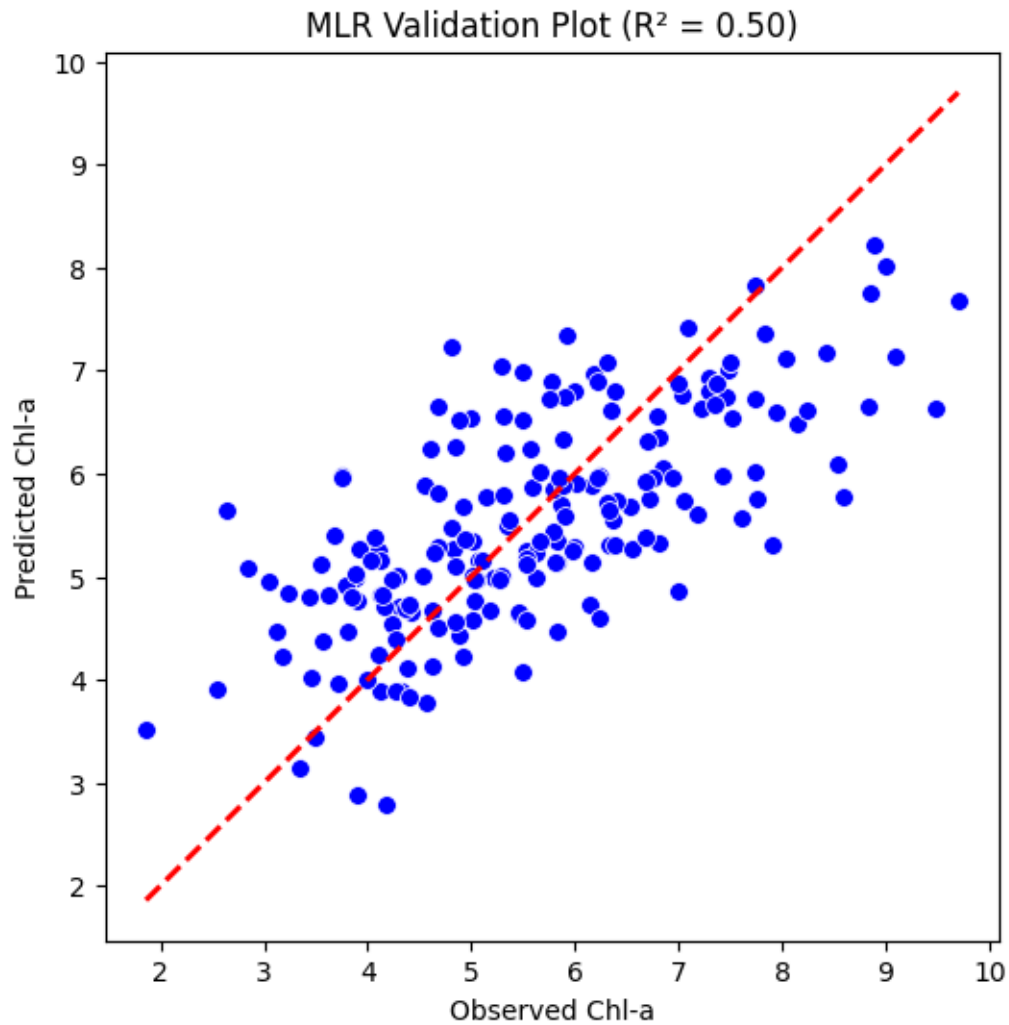
RMSE = 1.072

MAE = 0.863

```

[7]: plt.figure(figsize=(6,6))
sns.scatterplot(x=y_test, y=y_pred, s=50, color='blue')
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw=2)
plt.xlabel("Observed Chl-a")
plt.ylabel("Predicted Chl-a")
plt.title(f"MLR Validation Plot (R2 = {r2:.2f})")
plt.show()

```



```
[8]: coef_df = pd.DataFrame({
    'Feature': X_train.columns,
    'Coefficient': mlr_model.coef_
})
intercept = mlr_model.intercept_

print("\n Model Coefficients:")
display(coef_df)
print(f"Intercept: {intercept:.3f}")
```

Model Coefficients:

	Feature	Coefficient
0	temp	0.530897
1	salt	-0.172500

Intercept: -1.747

[]: